

Inverter-Based Power System Model for Benchmarking in Multi-Time-Scale Platforms

Manisha Maharjan¹, Jinho Kim¹, Sangwon Seo², Shrirang Abhyankar¹, Quan Nguyen¹, Yousu Chen¹, Nader Samaan¹, Eduard Muljadi², Sohom Datta¹, Bharat Vyakaranam¹

¹*Pacific Northwest National Laboratory, Richland, WA 99354, USA*

²*Auburn University, AL, 36849, USA*

Abstract—As the inverter-based resources (IBR) penetration increases, the approaches for conventional power system dynamics analysis will be inadequate for an IBR-dominated grid. The preliminary step for modernizing these analytical approaches is building a benchmark test system for both steady-state and dynamic studies for power grids with high penetration of IBRs while incorporating new sets of grid stability and reliability concerns. However, there is no such consistent test system available for IBR studies because individual researchers utilize different systems. To bridge the gap, this paper benchmarks the IEEE 39-bus test system with IBRs, by adjusting various model parameters based on recent industry practices and integrating with diversified IBRs to reflect accurate grid characteristics for power system phasor-based dynamics and electromagnetic transient studies. The simulation results are validated using several commercial tools. The potential grid services that can be offered by IBRs are also demonstrated in the test system. This test system can serve as a benchmark system for researchers to develop and validate new analytical approaches for steady-state, dynamic simulation and protection needed for reliability analysis of the future power grid. The test system data and various case studies built in different commercial tools are made publicly available for researchers' further utilization.

I. INTRODUCTION

Recent introduction of power electronic devices for inverter-based resources (IBRs) and reduction in system inertia, conventionally provided by synchronous machines, have raised numerous challenges for the modern power grid. Such challenges were recently witnessed in the Western Interconnection that resulted in the loss of thousands of MW of solar generation due to erroneous calculations of frequency using instantaneous frequency measurements [1], [2]. Such events highlight the necessity for detailed interconnection studies with IBRs and accurate representation of IBRs with improved analytical approaches. Current analytical approaches to simulate the behavior of the power grid are inadequate to consider the fluctuating dynamics of the power grid, which consists of synchronous machines and inertia-less IBRs. Thus, we need to advance the analytical approaches appropriate for this heterogeneous grid to understand the contributions of IBRs in system dynamics like inertial and governor responses. In this context, a reliable test system representing heterogeneous generation is essential for designing and validating new

analytical approaches to better understand the system dynamics with high penetration of IBRs. Unfortunately, there is no available test case with both synchronous and IBR generation that can be used for both phasor dynamics and electro-magnetic transient (EMT) studies.

The IEEE 39-bus system is widely used by researchers to represent a bulk power system. The generators and control systems for the test system are carefully chosen in a way to represent the actual machines connected to the New-England 345 kV system. However, there are several versions of this test system in the literature used for power system research. These test systems have different system parameters, component models, and responses built in different power system simulation tools. Using such variegated forms of the test system results in strenuous and erroneous efforts while comparing newly developed technologies, controllers, and solutions. A realistic and accurate effort is required to benchmark such test systems before any significant outcomes are considered from these variegated models used for representing the power grid in numerous studies. We also need these reliable system models in multiple simulation platforms to facilitate different types of power system studies and applications, as it might not be realized in a single power system tool. Only then can researchers make informed decisions about recognizing the best controls and solutions that fit ongoing challenges. This work constitutes rigorous validation and comparison of steady-state and dynamic responses of different publicly available variations of the IEEE 39-bus test system with IBRs in various power system simulation tools—available in [3]—to identify an accurate representation of the system responses for power system studies.

II. DEVELOPMENT OF BENCHMARK TEST BEDS OF IEEE 39-BUS SYSTEM WITH INVERTER-BASED RESOURCES

This section describes the modifications done to the original IEEE 39-bus system for creating benchmark testbeds based on recent industry practices. These modifications are necessary to preserve compatible simulation results in different simulation platforms: Phasor-based, EMT-based, and hybrid simulation platforms.

A. Original IEEE 39-Bus System

The main components of the IEEE 39-bus system are generators, governors, exciters, transmission lines, and loads as shown in Fig. 1. The detailed parameters for the components are presented in [4]–[6] in per unit with respect to a 100 MVA base. Table I shows the generator parameters of the original IEEE 39-bus system.

B. Implementation of Benchmark Test Beds for IEEE 39-Bus System

This subsection explains required modifications from the original IEEE 39-bus system parameters shown in Table I and how to generate corresponding benchmark test beds in EMT and hybrid simulation platforms based on Fig. 2.

1) Modification 1: Machine Base

All the generators in the original 39-bus system have the same machine base (M_{base}) value of 100 MVA, which can cause erroneous simulation results especially when power generated is greater than M_{base} . Therefore, M_{base} of each generator should be modified to reflect the practical capacity of the machine by considering each generator's scheduled active power, reserve power, and available reactive power. Based on these, new M_{baseN} , can be obtained by

$$M_{baseN} = \begin{cases} \sqrt{P_G^2 + Q_{max}^2} & \text{if } P_{cmd} < (1 - M_R)\sqrt{P_G^2 + Q_{max}^2} \\ \sqrt{P_G^2 + Q_{max}^2}/(1 - M_R) & \text{Otherwise} \end{cases} \quad (1)$$

where P_G , Q_{max} , P_{cmd} , and M_R are the scheduled active power in the original IEEE 39 bus system [6], maximum reactive power of a machine, user-defined command power, and MVA reserve margin of the machine, respectively. Q_G can be used when Q_{max} is not available in (1).

In (1), if a generator can secure a user-defined reserve power—for example, 10% of capacity which means M_R is 0.1—at a given value of P_{cmd} , then the M_{baseN} is set to the magnitude of $P_G + jQ_{max}$. Otherwise, M_{baseN} is set to 110% of the magnitude to secure the user-defined reserve. Table II provides the list of the generators with their M_{base} and M_{baseN} obtained based on (1).

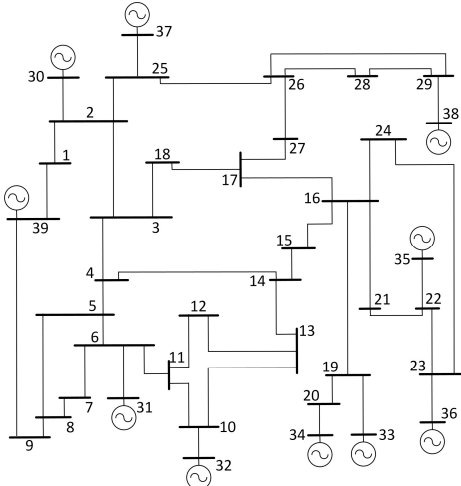


Fig. 1. Original IEEE 39-Bus System.

TABLE I. GENERATOR PARAMETERS OF THE IEEE 39-BUS SYSTEM [5]

Buses	P_G (MW)	Q_G (MVar)	Q_{max} (MVar)	M_{base} (MVA)	H (pu)
30	250	146.45	400	100	42.0
31	521	195.45	0.00	100	30.3
32	650	206.33	300	100	35.8
33	632	110.11	250	100	38.6
34	508	165.85	167	100	26.0
35	650	212.64	300	100	34.8
36	560	101.31	240	100	26.4
37	540	0.62	250	100	24.3
38	830	22.96	300	100	34.5
39	1000	88.96	-	100	500

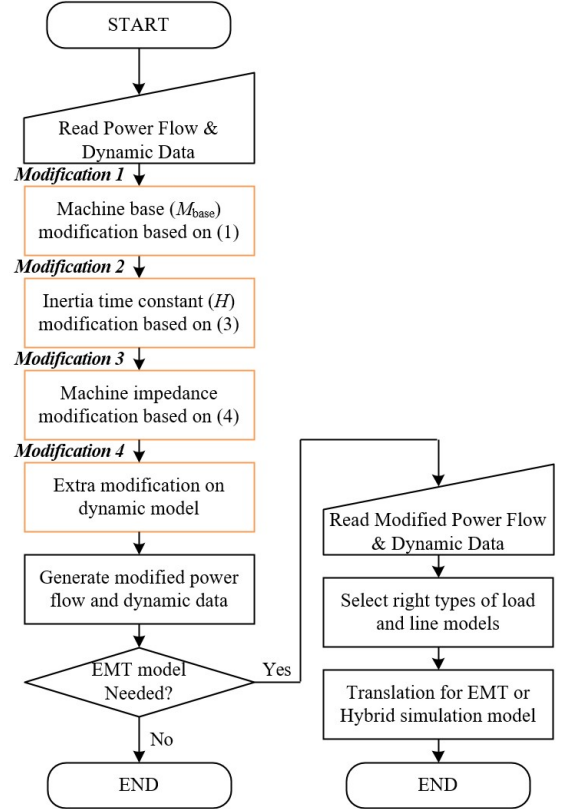


Fig. 2. Process of implementation of benchmark test beds.

2) Modification 2: Inertia Constant (H)

Original H values should be modified according to M_{base} modification. H is defined as

$$H = \frac{\frac{1}{2}J\omega^2}{M_{base}} \quad (2)$$

As seen in (2), assuming the moment of inertia, J , of the generator and nominal speed, ω , is constant, the time for a synchronous generator to produce M_{base} will be reduced at the same kinetic energy stored. Thus, new H , H_{New} , can be derived considering M_{baseN} by

$$H_{New} = H \times \frac{M_{base}}{M_{baseN}} \quad (3)$$

The change in the inertia constant with the modified M_{base} is listed in Table II.

3) Modification 3: Generator Impedance Parameters

Similarly, the impedance parameters of the generators should be modified with respect to the modification made in M_{base} . As the M_{base} value is modified, the per unit values for generator reactance are replaced respect to the new impedance base values for each generator using (4).

$$X_{New} = X \times \frac{M_{base}}{M_{baseN}} \quad (4)$$

Table III shows the modifications made in the impedance values of a generator at bus 30 as an example.

4) Modification 4: Additional modifications

Due to unavailability of some models among different simulation platforms, additional modifications might be needed to get compatible test beds. For example, the governor model named TGOV2 is available in PSSE but not in PSLF. In this case, TGOV1, shown in Fig. 3, can be used. The main components represented in this model are turbine-governor droop, R , main steam control valve limits, V_{max} and V_{min} , and time constants related to the motion of the steam in the turbine and heating stages. The maximum valve limits for TGOV1 in the test system are needed to be changed according to M_{baseN} .

Through the modification processes, the final outcomes—modified power flow and dynamic data—are obtained for the IEEE 39-bus system. The outcomes can be converted for power system engineers who are using different bulk power system simulation platforms.

5) Developing EMT or Hybrid Simulation Test Beds

In addition, the outcomes can be leveraged to obtain equivalent EMT domain or hybrid simulation model as described in Fig. 2.

In this paper, this effort was done by utilizing a commercial power system translation software, E-tran. E-tran can preserve the very similar machine dynamics by translating them based on the power flow and dynamic data provided and selecting right load and line models to be used in EMT simulation.

For the hybrid simulation test bed model, a user can define specific buses to translate them in EMT domain using E-tran. Then, E-tran sets the boundary buses where the EMT domain model communicate with the phasor domain model.

C. Modified IEEE 39-Bus System with Generic IBR Models

1) Generic Models of IBR

The WECC modeling validation subcommittee has been developing available set of generic IBR models for large system planning studies. The generic IBR model consists of the power plant controller (REPC), the renewable energy electrical control model (REEC), and generator/converter model (REGC). The detailed control block diagrams are found in [7].

2) Implementation of Generic IBR Models in PSCAD

Since the generic IBR models are not provided in EMT domain simulation, they should be implemented for benchmark test beds in EMT domain. Fig. 4 shows the overall configuration of the generic IBR model implemented in PSCAD. The REPC(A), REEC(A), and REGC(A) models are the same as in PSS/E and PSLF.

TABLE II. MODIFIED GENERATOR PARAMETERS FROM TABLE I

Buses	P_G (MW)	Q_G (MVar)	Q_{max} (MVar)	M_{base} (MVA)	M_{baseN} (MVA)	H (pu)	H_{New} (pu)
30	250	146.45	400	100	471	42.0	8.90
31	521	195.45	0.00	100	556	30.3	5.44
32	650	206.33	300	100	722	35.8	4.95
33	632	110.11	250	100	680	38.6	5.67
34	508	165.85	167	100	534	26.0	4.86
35	650	212.64	300	100	715	34.8	4.86
36	560	101.31	240	100	610	26.4	4.33
37	540	0.62	250	100	595	24.3	4.08
38	830	22.96	300	100	882	34.5	3.90
39	1000	88.96	-	100	10000	500	5

TABLE III. REACTANCE VALUES FOR GENERATOR 30

Generator parameters (pu)		X	X_{New}
X_d	d-axis synchronous reactance	0.100	0.0212
X_q	q-axis synchronous reactance	0.069	0.0146
X'_d	d-axis transient reactance	0.031	0.0066
X'_q	q-axis transient reactance	0.030	0.0064
X''_d	sub-transient reactance	0.020	0.0042
X_l	leakage inductance	0.0125	0.0027

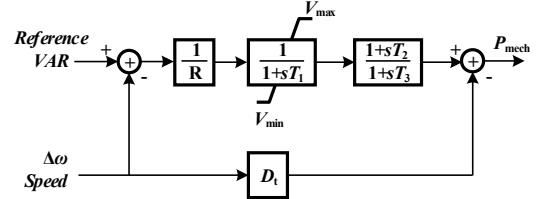


Fig. 3. Turbine-Governor System (TGOV1).

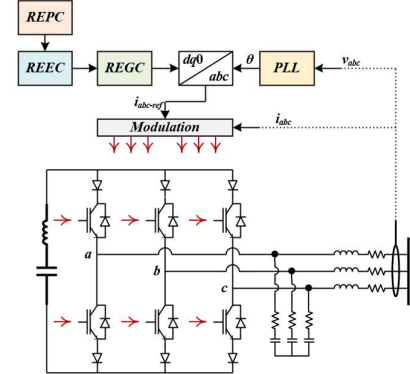


Fig. 4. Configuration of an IBR model implemented in PSCAD.

Phasor-locked loop (PLL) dynamics are included in the PSCAD implementation. The phase output of the PLL depends on the input 3-ph voltage, v_{abc} in Fig. 4, measured at the terminal. Thus, the performance of the PLL is highly affected by weak-grid connection and any voltage-related disturbances. Also, a power filter is required for the PSCAD implementation, and this might cause discrepancy with the source impedance in the phasor-domain models.

III. BENCHMARKING RESULTS AND USAGES OF THE DEVELOPED TEST BEDS

To validate the enhanced IEEE 39-bus IBR system modeled in different power system simulator tools, the steady-state and dynamic behaviors of the test system with and without IBRs are captured under different scenarios. The test results are divided into three parts: Benchmarking simulation results and usages of the developed test beds as described in Table IV.

TABLE IV. CASE STUDIES

	Description
Benchmarking results	Enhanced IEEE 39-bus system
	Enhanced IEEE 39-bus system with three IBRs
Usages of Test beds	Fast Frequency Response [8], [9]
	Offshore Wind Integration Study [10]
	Optimal Droop Setting for Congestion Reduction [11]
	Evaluation of Frequency Response of Power Grid [12]

A. Enhanced IEEE 39-Bus System

This section shows the responses from the proposed enhanced IEEE 39-bus systems in PSS/E, PSLF, and PSCAD.

1) System Response for a 3-ph Fault at Bus 16

A three-phase fault at bus 16 for 0.1 s was applied at 5 s in the test system. Fig. 5 and Fig. 6 show the active power output and voltage at the generator terminals. The steady-state power and voltage at these generator terminals before 5 s are same in all three platforms: PSS/E, PSLF, and PSCAD. During the fault, the power and voltage from the generators show similar dynamics and settle to the same value in the responses in all three platforms.

2) Frequency Response for Generator Trip

Similarly, to compare the responses from the test system with a frequency event, generator 32 is taken out of service after 5 s in all three simulation tools. Fig. 7 and Fig. 8 show the active power output and the frequency response from the generators. The responses from the generators show similar pattern and frequency dip in the generator responses. However, there are minor deviations in the transients and final values among the responses seen among different platforms.

B. Enhanced IEEE 39-Bus System with Three IBRs

In this case, the benchmark test beds include three IBR power plants connected to bus 30. The test beds are implemented in PSS/E, PSLF, PSCAD, Hybrid (PSS/E + PSCAD), and GridPACK—Open-source high-performance computing software.

Fig. 9 shows the responses from the test system with IBRs for a generator trip at bus 32. Fig. 10 shows the results from the IBR terminal during the event. These results show that the power and frequency responses from the generators are comparable in all these platforms. There are minor deviations in the frequency nadir among different tools, but they settle out to similar values. From these results, it can be said that the enhanced IEEE 39-bus system is appropriate for IBR studies—verified by the comparable responses in all these power system tools—and can be further used for developing various use cases for studies related to integration of IBRs.

B. Usages of Developed Test Beds [8], [9]

We leveraged the enhanced IEEE 39-bus system with generic IBRs to develop and verify fast frequency response (FFR) schemes in [8] and [9].

1) FFR₁ [8]

In this work, we implemented a FFR scheme in the generic IBR model. The detailed implementation can be found in [8]. The FFR scheme was tested using the enhanced IEEE 39-bus IBR system in PSCAD. The test was conducted by considering different penetration levels (PLs) of the IBRs as shown in Fig. 11.

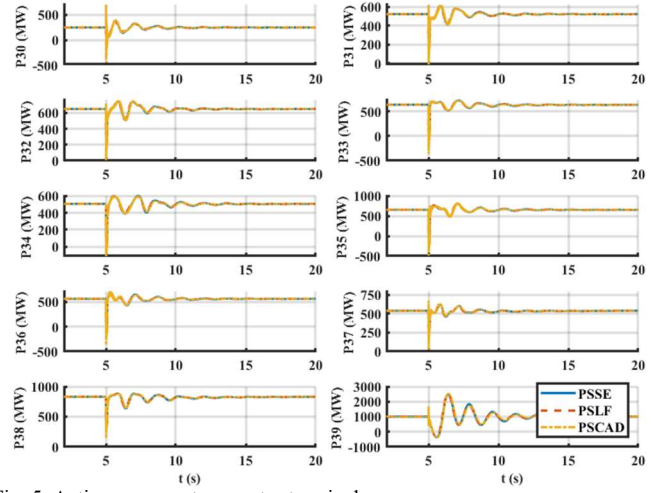


Fig. 5. Active powers at generator terminals.

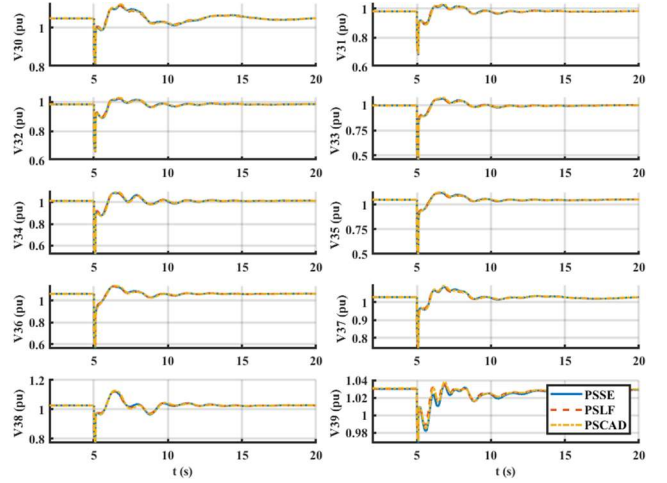


Fig. 6. Voltages at generator terminals.

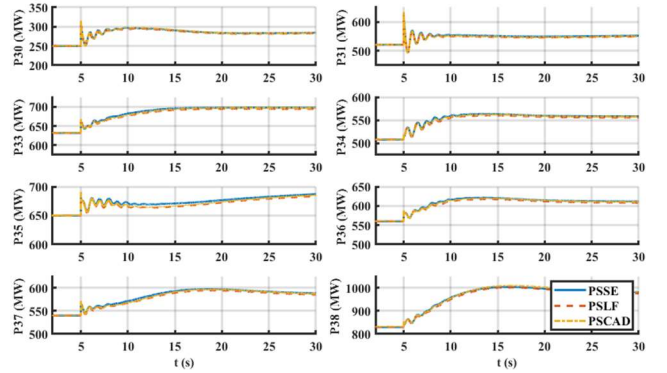


Fig. 7. Active powers at generator terminals.

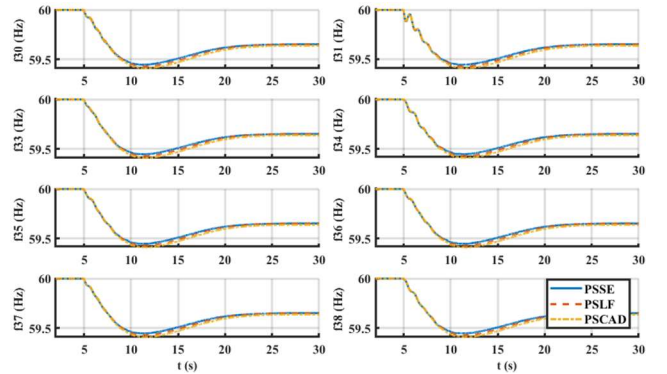


Fig. 8. Frequencies at generator terminals.

2) FFR₂ [9]

In this work, we implemented a quantified FFR scheme in the generic IBR model. The detailed implementation can be found in [9]. The proposed scheme was tested using the enhanced IEEE 39-bus IBR system. The efficacy of the proposed scheme was validated using PSLF and PSCAD as shown in Fig. 12.

V. CONCLUSIONS AND FUTURE WORK

This paper aims to improve the IEEE 39-bus system, allowing power system engineers to modernize their analytic methods for power system stability and control by incorporating different levels of IBRs into their analysis tools. Different benchmark models are generated using the multi-time scale enhanced IEEE 39-bus system and WECC-approved generic models of IBRs. These benchmark models are tested and verified in electromechanical and electromagnetic transient simulation platforms with comparisons of steady-state and dynamic behavior during contingencies. Further, the benchmark models are also simulated in a hybrid framework using electromechanical and electromagnetic simulation programs, and in the open-source high-performance computing tool GridPACK. This enhanced IEEE 39-bus IBR system is made publicly available in [3] and will serve as benchmark system for developing and validating new analytical approaches for steady-state, dynamic simulation and protection needed for reliability analysis of the future power grid.

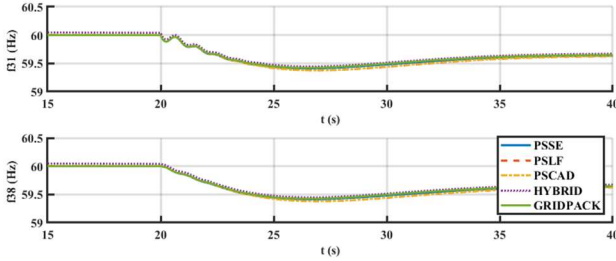


Fig. 9. Frequencies at generator terminals.

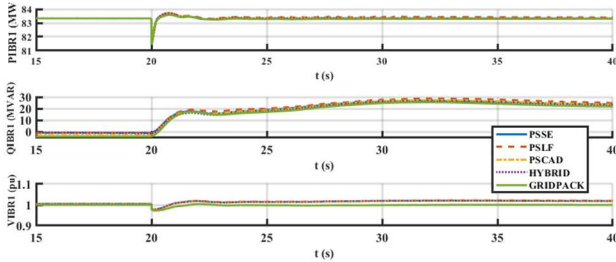


Fig. 10. Active power, reactive power, and voltage at IBR terminals.

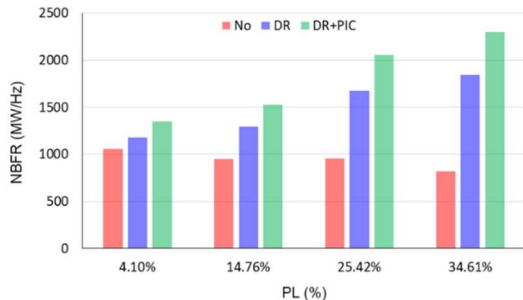


Fig. 11. Nadir-based frequency response (NBFR) considering different frequency controls using enhanced IEEE 39-bus system in PSCAD.

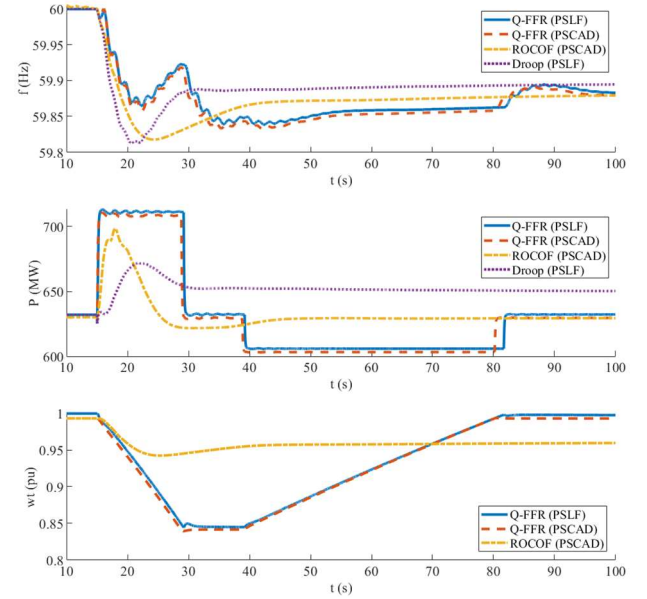


Fig. 12. Validation of quantified fast frequency response using the enhanced IEEE 39-bus system in PSLF and PSCAD.

REFERENCES

- [1] NERC, "900 MW Fault Induced Solar Photovoltaic Resource Interruption Disturbance Report: Southern California Event: October 9, 2017 Joint NERC and WECC Staff Report," ed: NERC Atlanta, GA, USA, 2018.
- [2] NERC, "1,200 MW Fault Induced Solar Photovoltaic Resource Interruption Disturbance Report: Southern California 8/16/2016 Event," ed: NERC, Atlanta, GA, USA, 2017.
- [3] Enhanced-IEEE-39-Bus-System-with-Inverter-based-Resources-on-Multi-Time-Scale-Platforms [Online] Available: <https://github.com/pnnltestsystem/Enhanced-IEEE-39-Bus-System-with-Inverter-based-Resources-on-Multi-Time-Scale-Platforms.git>
- [4] T. Athay, R. Podmore, and S. Virmani, "A practical method for the direct analysis of transient stability," *IEEE Transactions on Power Apparatus and Systems*, no. 2, pp. 573-584, 1979.
- [5] A. Pai, *Energy function analysis for power system stability*. Springer Science & Business Media, 1989.
- [6] E. Gulachenski and J. Undrill, "345kV Transmission System Typical of The New England Area," *Sample Power System Data for EEL Research Project on Bulk Power Transmission System Security (RP-90)*, pp. 20-35.
- [7] D. Ramasubramanian, "Model user guide for generic renewable energy system models," Electr. Power Res. Inst., Palo Alto, CA, USA, Tech. Rep. 3002027129, 2023.
- [8] J. Kim et al., "Enhanced Frequency Support Scheme of Generic Inverter-Based Resource Models for Renewable-Dominated Power Grids," *2022 IEEE Energy Conversion Congress and Exposition (ECCE)*, Detroit, MI, USA, 2022, pp. 1-7.
- [9] J. Kim et al., "Quantified Fast Frequency Response of Inverter-Based Resources for Bulk-Power System Study," *2024 IEEE Power & Energy Society General Meeting (PESGM)*, Seattle, WA, USA, 2024, pp. 1-5.
- [10] M. Maharjan et al., "EMT and Phasor-Domain Co-simulation of a Low Frequency AC Transmission System for Offshore Wind Integration," *2023 IEEE Power & Energy Society General Meeting (PESGM)*, Orlando, FL, USA, 2023, pp. 1-5.
- [11] Q. Nguyen et al., "Optimal Droop Setting for Congestion Reduction in a 100% Grid-Forming Inverter-based Power System," *2024 IEEE Power & Energy Society General Meeting (PESGM)*, Seattle, WA, USA, 2024, pp. 1-5.
- [12] S. Seo et al., "Evaluation of Primary Drivers Impacting Frequency Response of Power Grid Under Different IBR Penetration Levels," *2024 IEEE Power & Energy Society General Meeting (PESGM)*, Seattle, WA, USA, 2024, pp. 1-5.